

114 學年度
國立中山大學

課程名稱

硬體描述語言

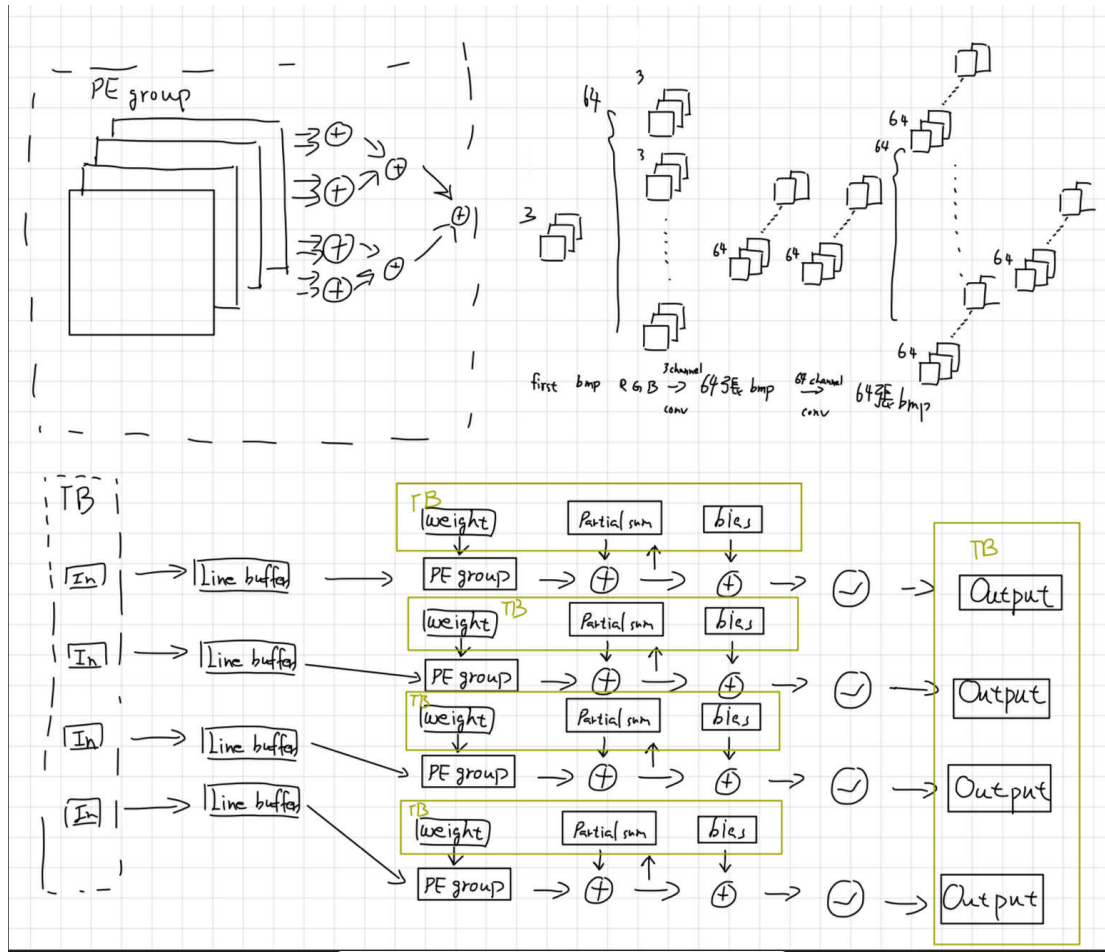
題目

HW5

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1.硬體架構解釋



Testbench 會將 bmp 的 img_data 傳入到 line buffer 裡面等到 linebuffer 湊滿捲機的條件後便會把計算出 r0~r8 傳入 PEgroup，pegroup 則負責做捲機及加法的部分並且傳出 partial sum 回給 testbench，重複運算直到滿足輸出條件，便會執行 relu 判斷輸出，layer1 和 layer2 的差別在於 layer1 不會用到 4 個 buffer 只會用到三個並且 kernal 深度也沒用到 64 個。

2. area power and critical path

Critical path:

L2_inst/LB0/r372/U335/Z	(OR2D1BWP16P90LVT)	0.01	3.11 f
L2_inst/LB0/r372/U263/Z	(OR2D1BWP16P90LVT)	0.01	3.12 f
L2_inst/LB0/r372/U192/Z	(AN2D1BWP16P90LVT)	0.01	3.13 f
L2_inst/LB0/r372/U128/Z	(AN2D1BWP16P90LVT)	0.01	3.15 f
L2_inst/LB0/r372/U77/Z	(ND2D1BWP16P90LVT)	0.01	3.15 r
L2_inst/LB0/r372/U61/Z	(INR2D1BWP16P90LVT)	0.01	3.16 r
L2_inst/LB0/r372/U20/Z	(CKND1BWP16P90LVT)	0.03	3.19 f
L2_inst/LB0/r372/U539/Z	(MUX2ND2BWP16P90LVT)	0.03	3.22 r
L2_inst/LB0/r372/U505/Z	(CKND1BWP16P90LVT)	0.01	3.23 f
L2_inst/LB0/r372/U441/Z	(OR2D1BWP16P90LVT)	0.01	3.25 f
L2_inst/LB0/r372/U336/Z	(OR2D1BWP16P90LVT)	0.01	3.26 f
L2_inst/LB0/r372/U264/Z	(OR2D1BWP16P90LVT)	0.01	3.27 f
L2_inst/LB0/r372/U183/Z	(AN2D1BWP16P90LVT)	0.01	3.29 f
L2_inst/LB0/r372/U120/Z	(AN2D1BWP16P90LVT)	0.01	3.30 f
L2_inst/LB0/r372/U78/Z	(ND2D1BWP16P90LVT)	0.01	3.30 r
L2_inst/LB0/r372/U62/Z	(INR2D1BWP16P90LVT)	0.01	3.32 r
L2_inst/LB0/r372/U21/Z	(CKND1BWP16P90LVT)	0.03	3.35 f
L2_inst/LB0/r372/U540/Z	(MUX2ND2BWP16P90LVT)	0.03	3.37 r
L2_inst/LB0/r372/U507/Z	(CKND1BWP16P90LVT)	0.01	3.38 f
L2_inst/LB0/r372/U440/Z	(OR2D1BWP16P90LVT)	0.01	3.40 f
L2_inst/LB0/r372/U338/Z	(OR2D1BWP16P90LVT)	0.01	3.41 f
L2_inst/LB0/r372/U265/Z	(OR2D1BWP16P90LVT)	0.01	3.43 f
L2_inst/LB0/r372/U193/Z	(AN2D1BWP16P90LVT)	0.01	3.44 f
L2_inst/LB0/r372/U121/Z	(AN2D1BWP16P90LVT)	0.01	3.45 f
L2_inst/LB0/r372/U79/Z	(ND2D1BWP16P90LVT)	0.01	3.46 r
L2_inst/LB0/r372/U88/Z	(INR2D1BWP16P90LVT)	0.01	3.47 r
L2_inst/LB0/r372/U22/Z	(CKND1BWP16P90LVT)	0.03	3.50 f
L2_inst/LB0/r372/U541/Z	(MUX2ND2BWP16P90LVT)	0.03	3.53 r
L2_inst/LB0/r372/U508/Z	(CKND1BWP16P90LVT)	0.01	3.54 f
L2_inst/LB0/r372/U442/Z	(OR2D1BWP16P90LVT)	0.01	3.55 f
L2_inst/LB0/r372/U337/Z	(OR2D1BWP16P90LVT)	0.01	3.56 f
L2_inst/LB0/r372/U266/Z	(OR2D1BWP16P90LVT)	0.01	3.58 f
L2_inst/LB0/r372/U194/Z	(AN2D1BWP16P90LVT)	0.01	3.59 f
L2_inst/LB0/r372/U129/Z	(AN2D1BWP16P90LVT)	0.01	3.60 f
L2_inst/LB0/r372/U80/Z	(ND2D1BWP16P90LVT)	0.01	3.61 r
L2_inst/LB0/r372/U89/Z	(INR2D1BWP16P90LVT)	0.01	3.62 r
L2_inst/LB0/r372/U23/Z	(CKND1BWP16P90LVT)	0.03	3.65 f
L2_inst/LB0/r372/U25/Z	(MUX2ND2BWP16P90LVT)	0.03	3.68 f
L2_inst/LB0/r372/U485/Z	(OR2D1BWP16P90LVT)	0.01	3.70 f
L2_inst/LB0/r372/U449/Z	(OR2D1BWP16P90LVT)	0.01	3.71 f
L2_inst/LB0/r372/U341/Z	(OR2D1BWP16P90LVT)	0.01	3.73 f
L2_inst/LB0/r372/U244/Z	(AN2D1BWP16P90LVT)	0.01	3.74 f
L2_inst/LB0/r372/U195/Z	(AN2D1BWP16P90LVT)	0.01	3.75 f
L2_inst/LB0/r372/U130/Z	(ND2D1BWP16P90LVT)	0.01	3.76 r
L2_inst/LB0/r372/U81/Z	(INR2D1BWP16P90LVT)	0.01	3.77 r
L2_inst/LB0/r372/U24/Z	(CKND1BWP16P90LVT)	0.03	3.79 f
L2_inst/LB0/r372/U682/Z	(MUX2D1BWP16P90LVT)	0.02	3.82 r
L2_inst/LB0/r372/remainder[5] (Line_buffer_4_DW_div_uns_6)		0.00	3.82 r
L2_inst/LB0/U26/Z (OR4D1BWP16P90LVT)		0.01	3.83 r
L2_inst/LB0/U33/Z (NR4D1BWP16P90LVT)		0.01	3.84 f
L2_inst/LB0/U34/Z (NR4D1BWP16P90LVT)		0.09	3.93 r
L2_inst/LB0/enable (Line_buffer_4)		0.00	3.93 r
L2_inst/C00/enable (conv_3x3_ct_16)		0.00	3.93 r
L2_inst/C00/U33/Z (INR2D1BWP16P90LVT)		0.10	4.03 r
L2_inst/C00/U3/Z (AN2D1BWP16P90LVT)		0.04	4.07 r
L2_inst/C00/partial_sum_reg_35_7D (DFQD2BWP16P90LVT)			
data arrival time		0.00	4.07 r
clock clk (rise edge)	10.00	10.00	
clock network delay (ideal)	0.00	10.00	
L2_inst/C00/partial_sum_reg_35_7CP (DFQD2BWP16P90LVT)			
library setup time	0.00	10.00 r	
data required time	-0.02	9.98	
data required time		9.98	
data arrival time		-4.07	
slack (MET)		5.91	

area:

```
*****
Report : area
Design : vgg16
Version: I-2022.03
Date : Thu Nov 27 23:59:15 2025
*****

Library(s) Used:

N16ADFP_StdCells0p72vm40c_ccs (File: /cad/CSDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCells0p72vm40c_ccs.db)

Number of ports: 45800
Number of nets: 202774
Number of cells: 133974
Number of combinational cells: 103594
Number of sequential cells: 29871
Number of macros/black boxes: 0
Number of buf/inv: 4128
Number of references: 9

Combinational area: 51142.805170
Buf/inv area: 664.796186
Noncombinational area: 27873.227887
Macro/black box area: 0.000000
Net Interconnect area: undefined (Wire load has zero net area)

Total cell area: 79016.033057
Total area: undefined

Hierarchical area distribution
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power:

```
*****
Report : power
        -analysis_effort low
Design : vgg16
Version: T:2022.03
Date   : Thu Nov 27 23:59:40 2025
*****

Library(s) Used:
    N16ADFP_StdCellss0p72vm40c_ccs (File: /cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCellss0p72vm40c_ccs.db)

Operating Conditions: ss0p72vm40c  Library: N16ADFP_StdCellss0p72vm40c_ccs
Wire Load Model Mode: top

Design      Wire Load Model      Library
-----
vgg16       ZeroWireLoad         N16ADFP_StdCellss0p72vm40c_ccs

Global Operating Voltage = 0.72
Power-specific unit information :
  Voltage Units = 1V
  Capacitance Units = 1.000000pf
  Time Units = 1ns
  Dynamic Power Units = 1mW (derived from V,C,T units)
  Leakage Power Units = 1nW

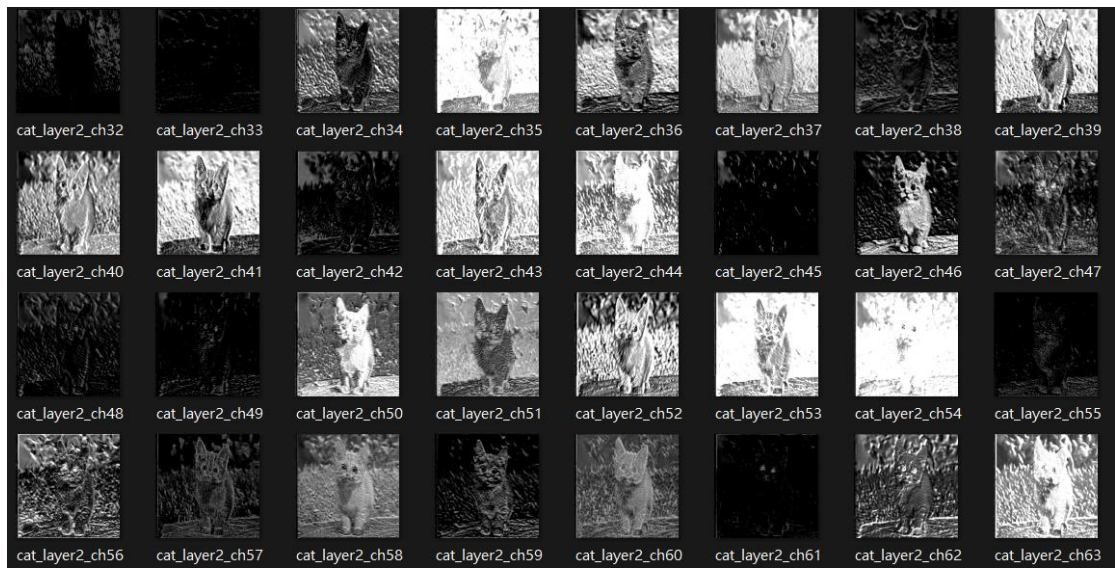
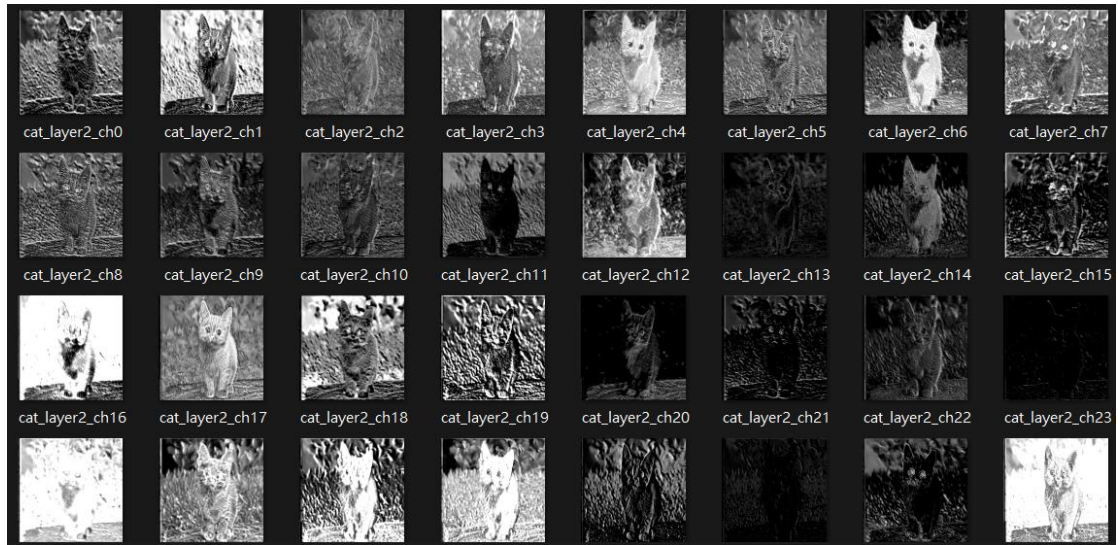
Attributes
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ti - Including register clock pin internal power

Cell Internal Power = 7.1871 mW (94%)
Net Switching Power = 445.2353 uW (6%)
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Total Dynamic Power = 7.6324 mW (100%)
Cell Leakage Power  = 45.4914 uW
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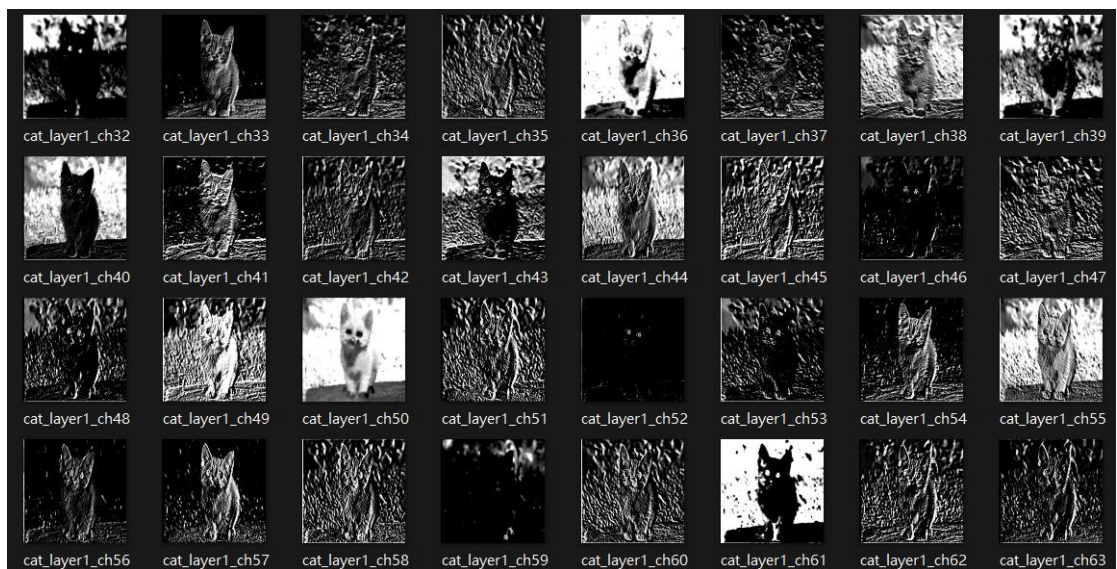
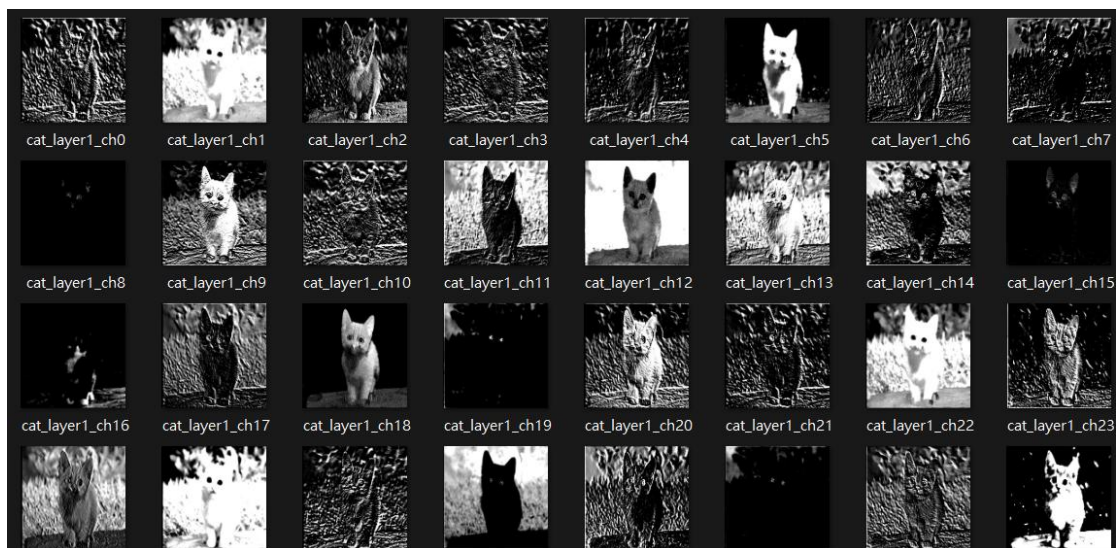
3. presim 輸出照片(conv1)



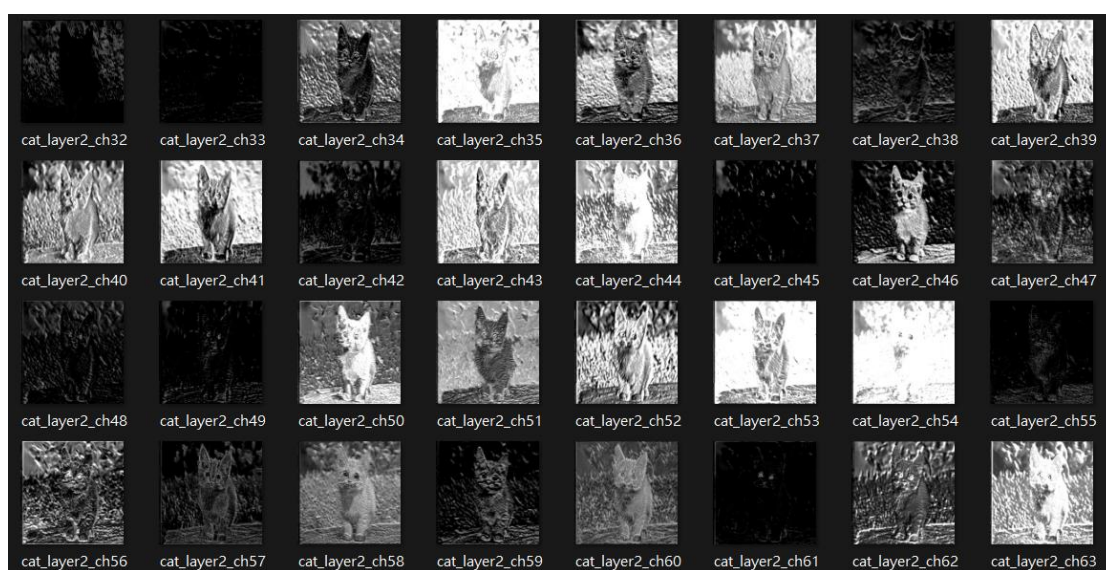
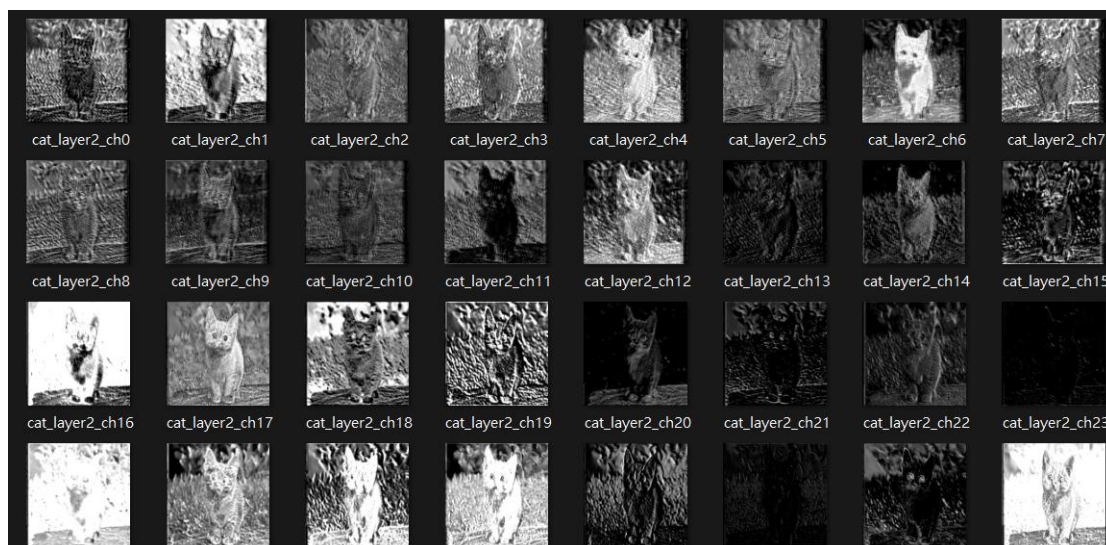
4. presim 輸出照片(conv2)



5. postsim 輸出照片(conv1)



6. postsim 輸出照片(conv2)



心得:

這次的 vgg16 滿好玩的，雖然說明看了很久才知道怎麼做但在發現其中的脈絡之後便可以快速的實作 layer1 跟 layer2 了，還有模擬的時候圖片也跑太久了吧，我 layer2 跑了快整整兩個小時阿，每次 debug 要等跑一張圖也要五分鐘，也太麻煩了吧，但是做出來看到圖片的時候就是超爽的，到了 postsim 就跑更久了從來沒跑過這麼久的 testbench。